A TEXMACS graphics tutorial

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1 Introduction

This document describes the graphical tool of TEXMACS, which you can use to make drawings, illustrations, diagrams, etc. In section 1.3, we introduce this graphical tool using a simple example, for which we show in detail the operations that a user must perform to build it step by step. In sections 2 to 8, we describe the available features of the graphical tool, and section 9 lists what changed since the first version of this tutorial. Finally, appendices A and B summarize the graphical markup of TEXMACS and the use of the SCHEME scripting language to generate calculated drawings.

The pictures of this tutorial were made with TEXMACS 2.1.5 (the TEXMACS Fixes application for MACOS); on other systems and with other looks, the toolbars look slightly different, and the keyboard shortcuts are those shown by the menus of your TEXMACS.

1.1 Overview

The graphical tool of TEXMACS is made of several complementary tools:

i. A *grid*, which helps you to position the objects precisely. You can change the parameters of the grid to make it finer or coarser, and change the colors of its different kinds of lines (axes, units, subunits). The available grids are *Cartesian*, *polar*, *logarithmic* and *notebook* grids (see section 2). Points are *snapped* to the grid, to the points and intersections of the existing curves and to the borders of the text boxes; what is snapped to is set in Focus→Snap.

ii. Tools for drawing *curves*: *points*, *lines*, *polygons*, *splines*, *smooth curves* and *Bezier curves* with their closed variants, *arcs* and *circles* (see section 3).

iii. *Text boxes*, *mathematical formulas* and *long text* boxes can be placed anywhere in the picture, with any TEXMACS markup inside; you can choose their horizontal and vertical alignment with respect to the anchor point which places them in the picture (see section 4). Hand-drawn curves can be added with the *penscript* and *calligraphy* tools.

iv. The *properties* of the objects: the *color*, *fill color*, *opacity*, *line width*, *dash style*, *arrows*, *point style* and the *alignment* of text boxes (see section 5).

v. You can *select* one or more objects, and *copy*, *cut* and *paste* them (see section 6).

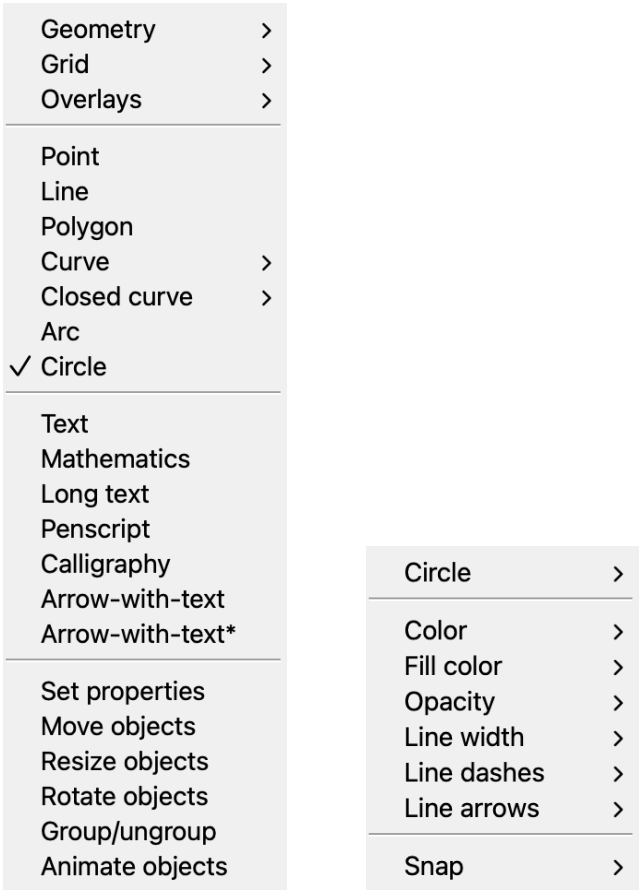
vi. Operations on a set of selected objects: *grouping and ungrouping*, *moving*, *resizing* and *rotating* (see section 7).

vii. Miscellaneous: changing the size of the picture, cropping it, moving the origin of its frame

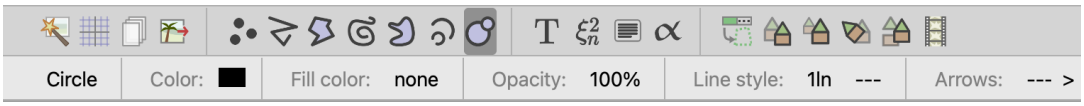
and zooming the whole picture (see section 8).

1.2 The user interface

When the cursor is inside a picture, the Insert menu becomes the menu of the graphical tool (on the left below), and the Focus menu gives the properties of the objects which will be drawn, for the current mode (on the right, for circles):



The mode dependent toolbar becomes the *graphics toolbar*:



Its first row is made of four groups of icons:

- The *global icons* give the Insert→Geometry menu (size, crop, alignment, unit, origin and zoom of the picture), the Insert→Grid menu, the overlays of the picture, and the icon to leave graphics mode:



- The *insertion modes* for curves: point, line, polygon, spline, closed spline, arc and circle (in the Insert menu: Point, Line, Polygon, Curve, Closed curve, Arc, Circle):



- The insertion modes for Text, Mathematics, Long text and hand-drawn curves:



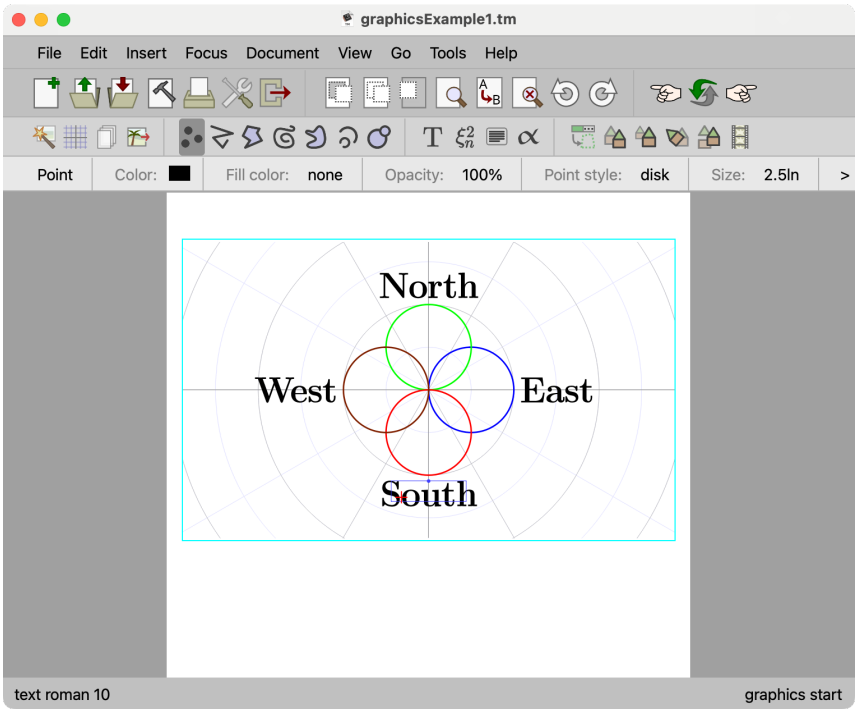
- The *group operations*: Set properties, Move objects, Resize objects, Rotate objects, Group/ungroup and Animate objects:



The second row, the *focus bar*, shows the current mode and the current values of the properties of the objects which will be drawn in this mode (color, fill color, opacity, line width and dashes, arrows...); click on a value to change it. For the group operations, it gives the snapping options.

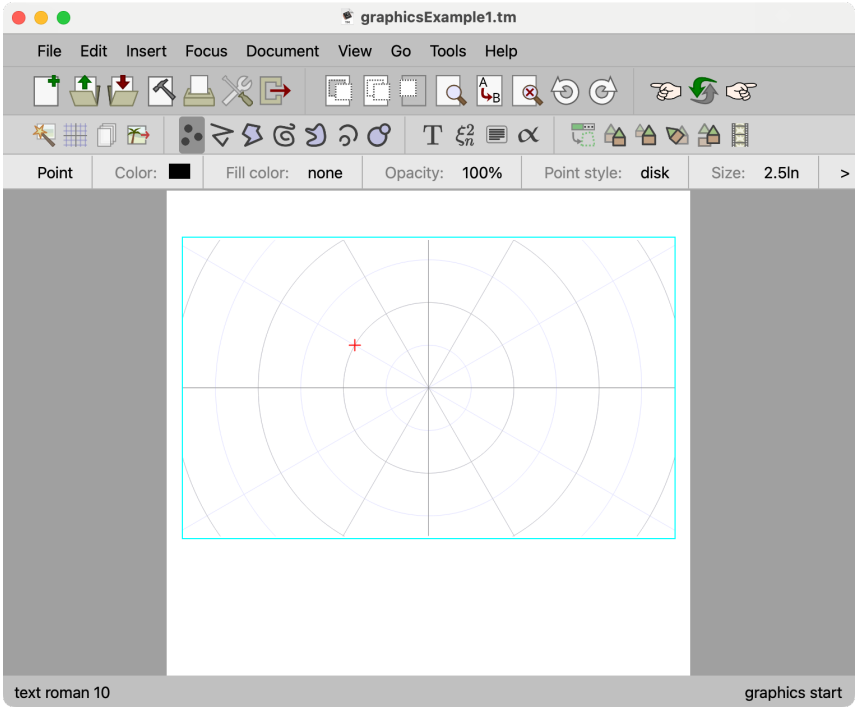
1.3 An introductory example

So that you can try the graphical tool of TEXMACS right away, even if you know nothing about it yet, we now show you step by step how to build the following drawing:



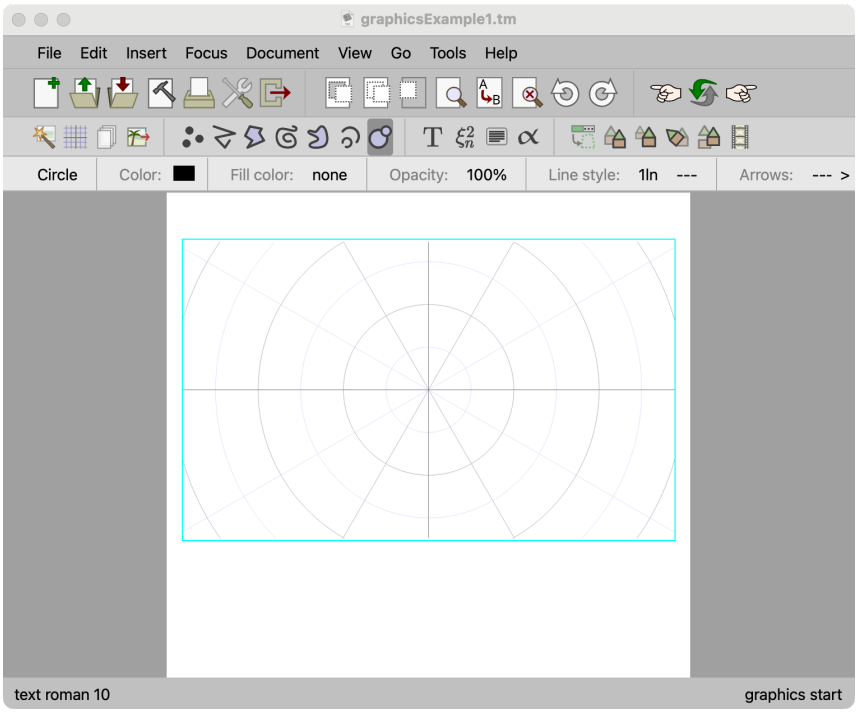
1.3.1 Insert a new picture and set its grid

First, insert a new picture in the document, using Insert→Image→Draw image. The cursor is now inside the picture, and the menus and toolbars are those of the graphical tool. Add a polar grid to the picture, using Insert→Grid→Type→Polar. Next, set the parameters of the grid: select Insert→Grid→Unit length→2, then Insert→Grid→Number of polar steps→6, then Insert→Grid→Number of subunit steps→2 (this entry appears when Insert→Grid→Show subunits is on, as it is by default). To see the whole picture at once, you can also reduce the zoom of the window (View→Zoom). You obtain the following grid:

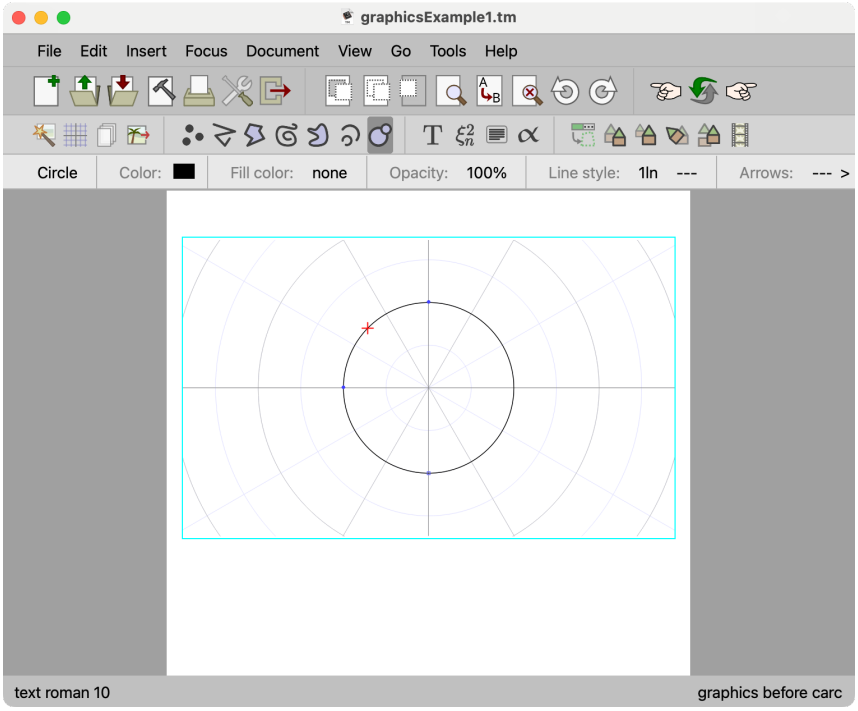


1.3.2 Insert a first circle

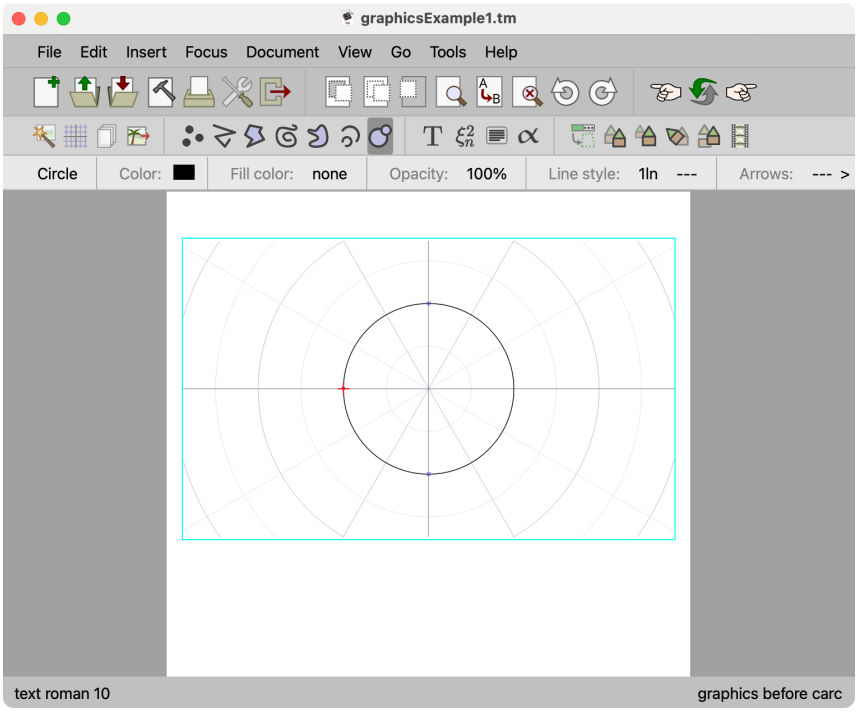
Now we insert a circle into the picture. Choose Insert→Circle (or the icon), move the mouse over the picture and click with the *left mouse button*. As soon as you move the mouse, you see that you are moving the end of a segment:



When you click again, the second point of the circle is added, and the segment becomes a moving circle through the two points and the mouse. Click a third time and then once more at the same place (a double click) to fix the last point: the whole circle is inserted in the picture, and you are ready to insert another object. The status bar always tells you what the mouse buttons do; for instance, while you are inserting the points, *Right click: undo* removes the last one.



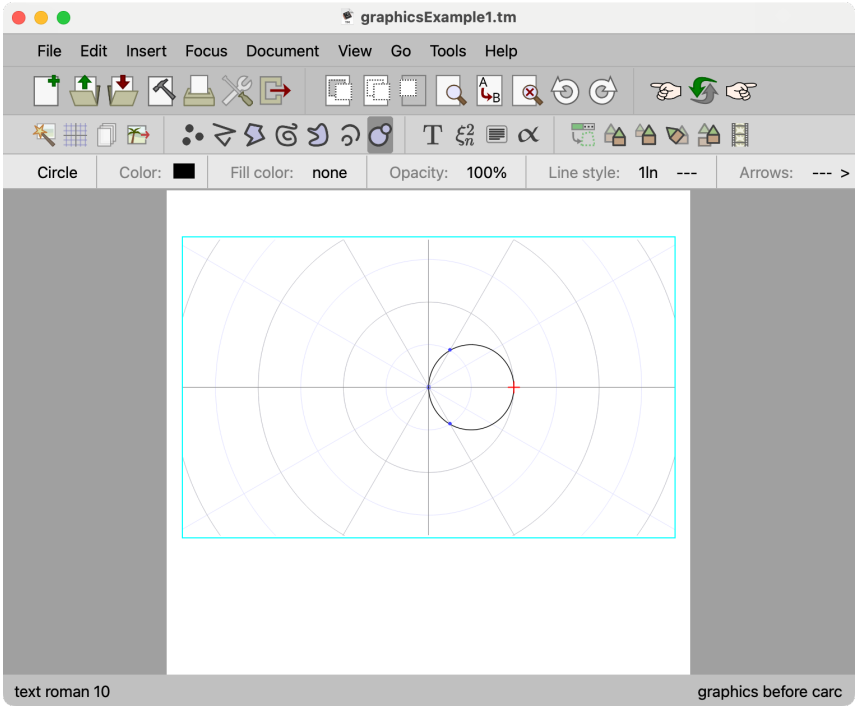
We now move the points of the circle to their intended positions. When the mouse is near the circle, its three *control points* are shown, and when the mouse is near one of them, this control point becomes the current one, and it is highlighted:



All curves (lines, splines, arcs...) are edited this way in TeXmacs:

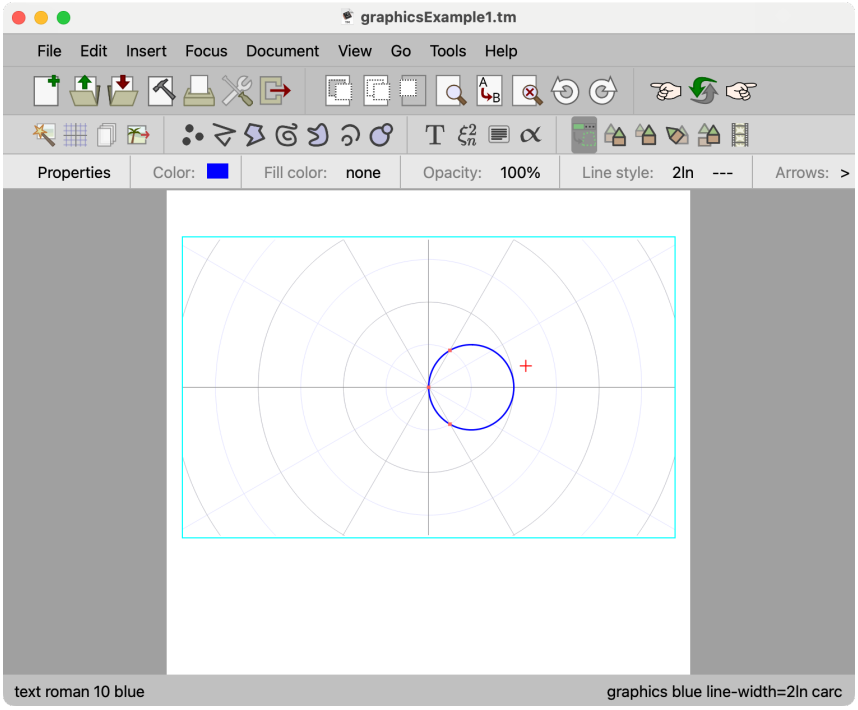
- to *move* a control point, drag and drop it with the *left mouse button*;
- when the mouse is on an *edge* of a curve with an arbitrary number of points (lines, polygons, splines), a drag and drop with the left mouse button *inserts* a new point there;
- the *middle mouse button* removes the control point under the mouse, and ⇧ with the middle mouse button removes the whole object;
- when several control points or objects are at the same place, the ⇥ key cycles through them.

Thanks to the snapping, the points fall exactly on the points of the grid. Move the three points of the circle to the origin and to the intersections of the first subunit circle with the rays at ±60 degrees. You obtain:



1.3.3 Set the properties of this first circle

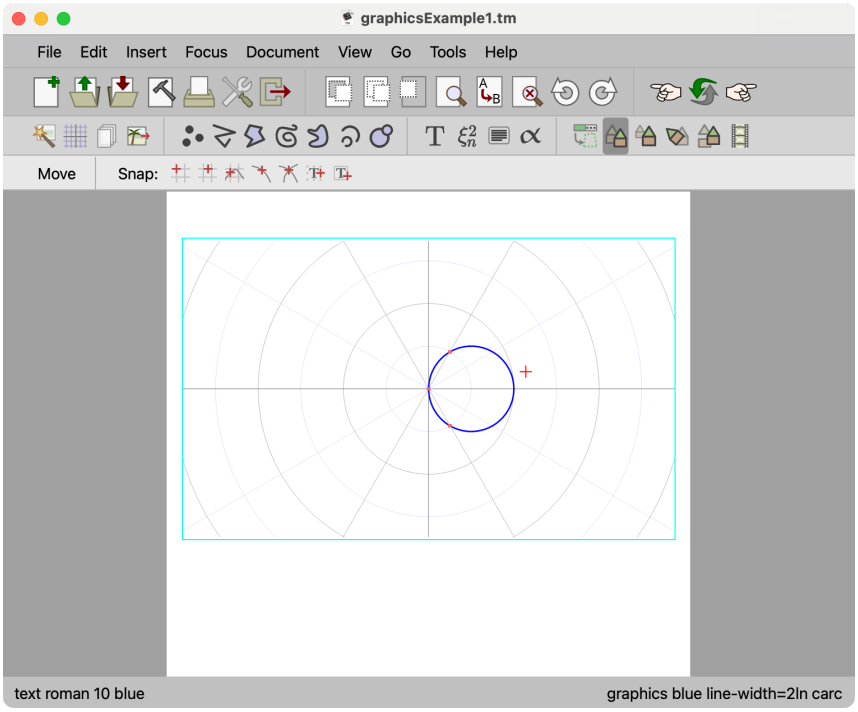
We now turn our circle into a thicker blue one. Select the Insert→Set properties mode (or the icon), and click on the circle with the left mouse button: the circle is selected (with the right mouse button, or by dragging a rectangle with it, you can select several objects), and the focus bar and the Focus menu now show its properties. Choose the color in Focus→Color→Blue and the width in Focus→Line width→2 ln (or in the focus bar): the change is applied at once to the selected circle, which becomes thicker and blue.



In the insertion modes, the Focus menu and the focus bar give the properties of the objects which will be drawn: if you choose a color and a width *before* drawing, the next objects are drawn with them. In any mode, ↩ applies these current properties to the object under the mouse.

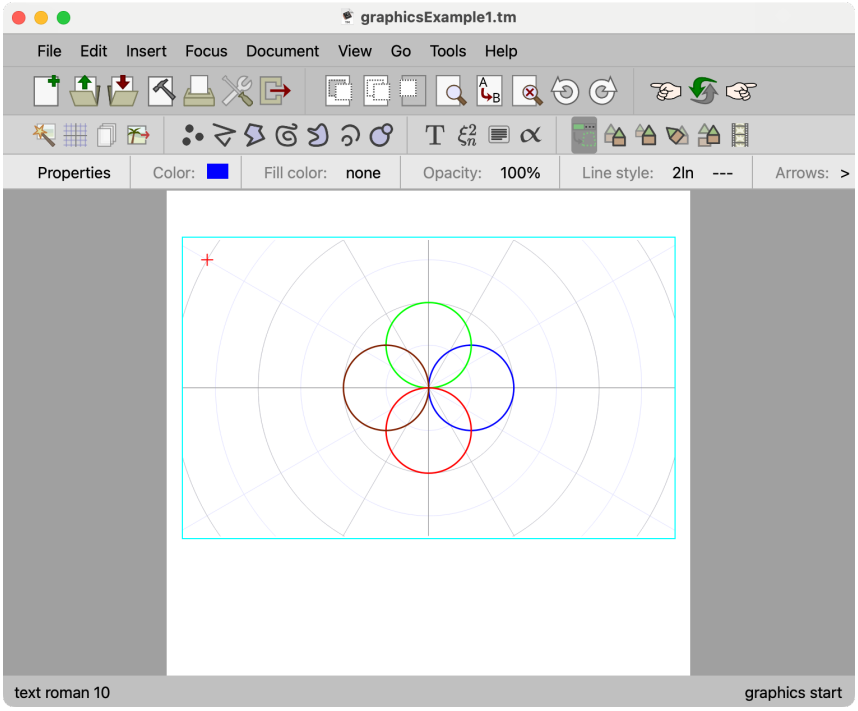
1.3.4 Make three copies of this circle

Now we see how to copy and paste graphical objects. Select the Insert→Move objects mode (icon), move the mouse over the circle and click with the *right mouse button*: the circle is selected. Then copy it with Edit→Copy (⌘C) and paste it with Edit→Paste (⌘V) three times: copy and paste work in the group modes, and the three copies lie on top of the original circle.



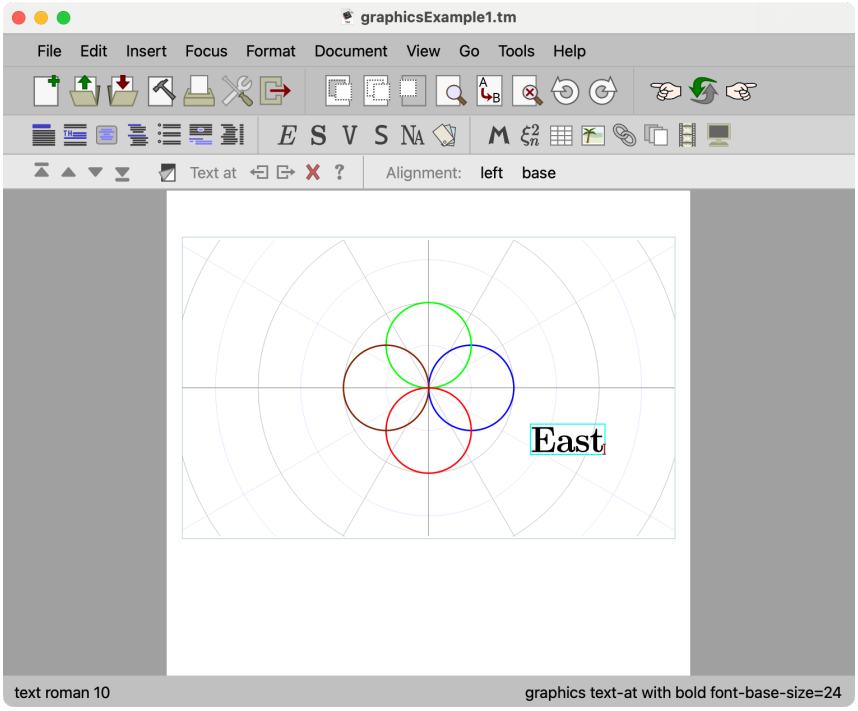
1.3.5 Set the position and the color of each of the circles

Now move each copy to its own position (the circles all go through the origin, and their centers are on the first subunit circle of the grid, to the north, the south and the west). In Insert→Move objects mode, a left click on a copy picks it up, and it follows the mouse until a second left click drops it (you can also drag and drop it). Then go back to the Insert→Set properties mode, click on a circle to select it and choose its color in Focus→Color: green for the north, red for the south and brown for the west. You obtain:



1.3.6 Add a text box

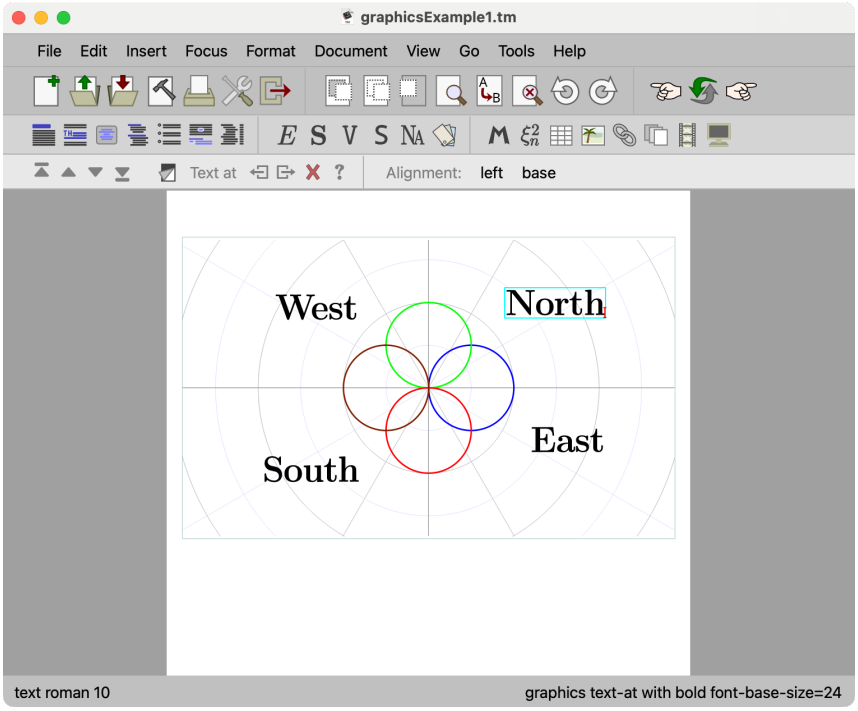
We now add a text box with the text "East", in bold face and in size 24. Choose Insert→Text (icon **T**), move the mouse into the picture and click with the left mouse button: a new text box is inserted, with the text cursor inside it. Choose the bold face and the size 24 (Format→Series→Bold, Format→Size→24, or the font toolbar), and type "East". You should obtain something like:



The toolbars are now those of the text inside the box, and the focus bar gives the alignment of the box. To leave the text box, click in the picture outside of it.

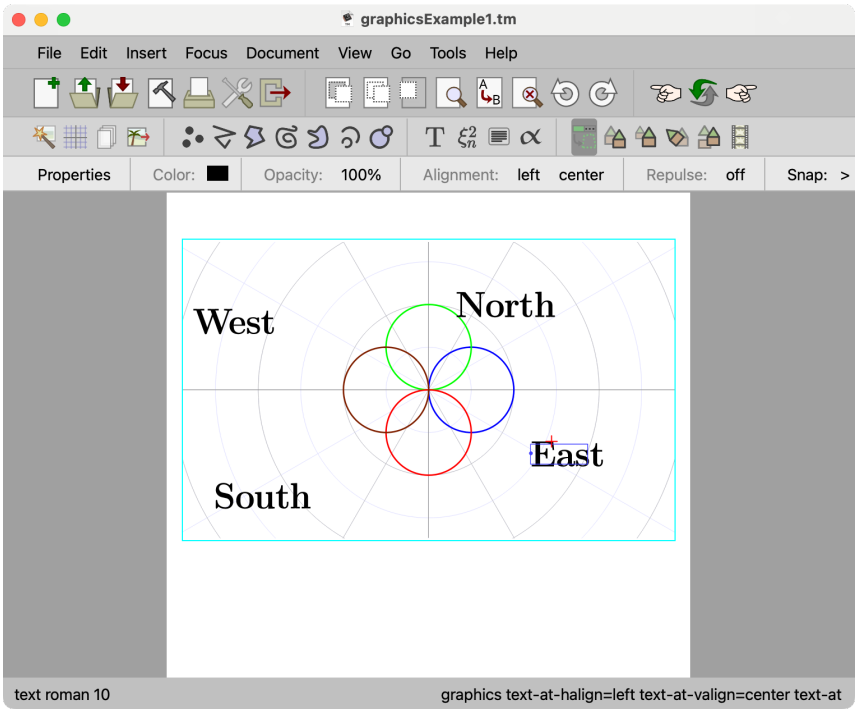
1.3.7 Make three copies of this text box

Now make three copies of the text box you just inserted, as you did for the circles. Move each box to a different position, then click into each copy to change its text. You get something like:



1.3.8 Change the alignment of the text boxes

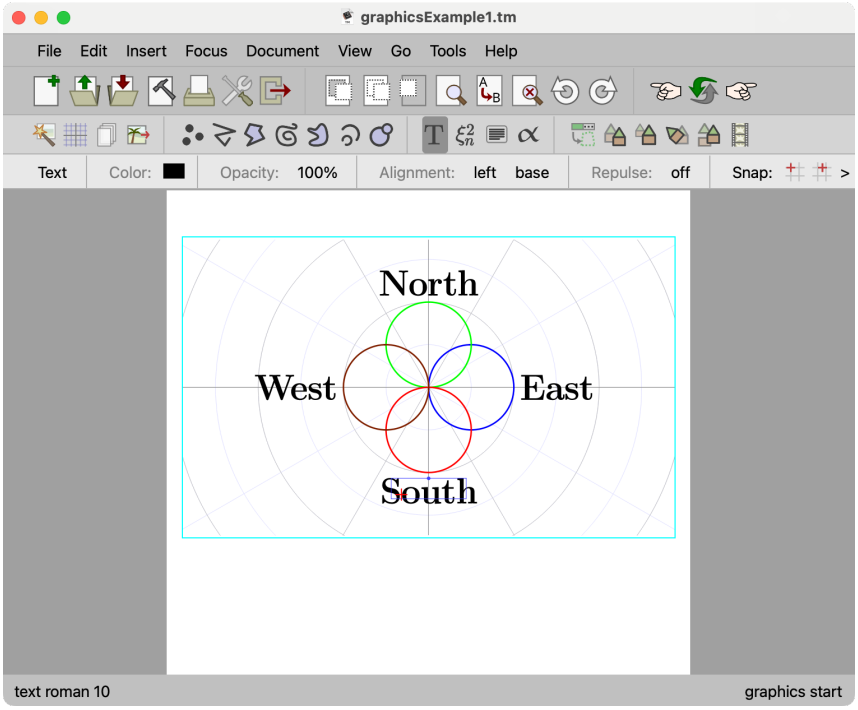
The position of a text box is given by an anchor point; its *horizontal alignment* (left, center or right) and its *vertical alignment* (bottom, base, axis, center or top) say how the box is placed with respect to this point. We give "West" a right alignment, "East" a left alignment, both centered vertically, and center "North" and "South" horizontally, "North" at the bottom and "South" at the top. To do this, choose the Insert→Set properties mode, click on a text box with the left mouse button to select it, and choose its alignment in Focus→Horizontal alignment and Focus→Vertical alignment (or in the focus bar). The boxes move with respect to their anchor points:



1.3.9 Move the text boxes to their intended positions

Now, in Insert→Text mode, drag and drop each text box (with the left mouse button) to its position:

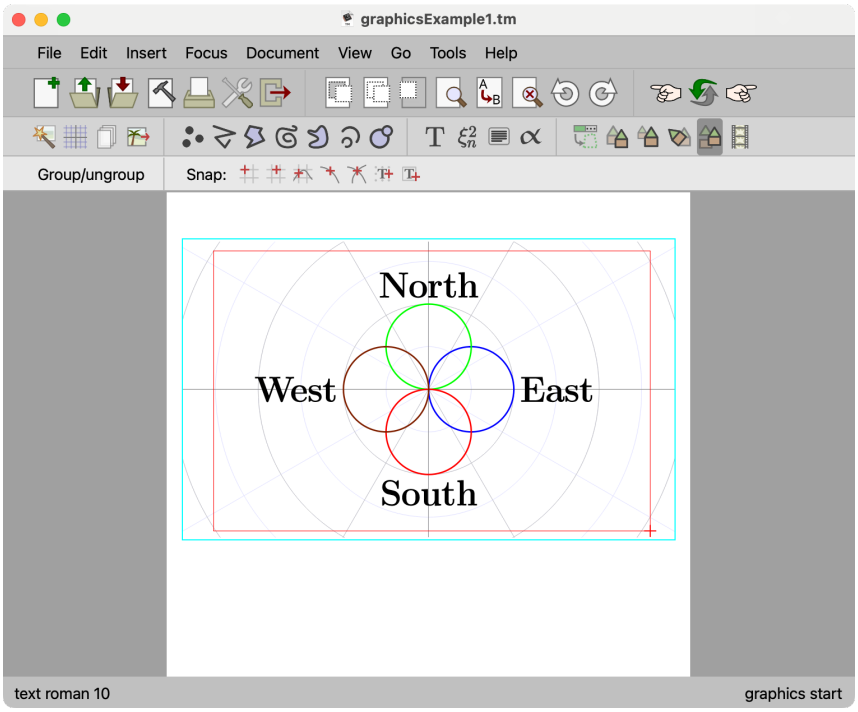
the anchor points snap to the points of the grid. Thanks to the alignments, each text sits next to its circle, outside of it:



To place the boxes more precisely, you can zoom in, with View→Zoom or with the keys of section 8, and come back to the usual zoom afterwards.

1.3.10 Select all, group it

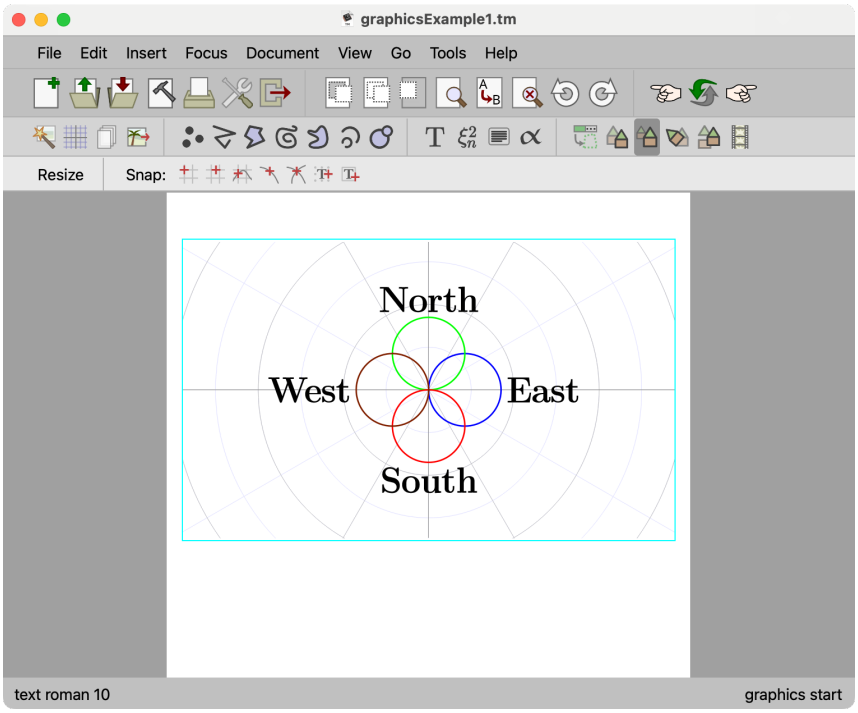
To do this, select the Insert→Group/ungroup mode (icon). You can then select all the objects at once: drag the *right mouse button* from one corner of a rectangle around the objects to the opposite corner; all the objects inside the rectangle are selected. You can also select the objects one after the other with right clicks (a right click on a selected object unselects it). Once the selection is done, click with the *left mouse button* to group the selected objects into one object:



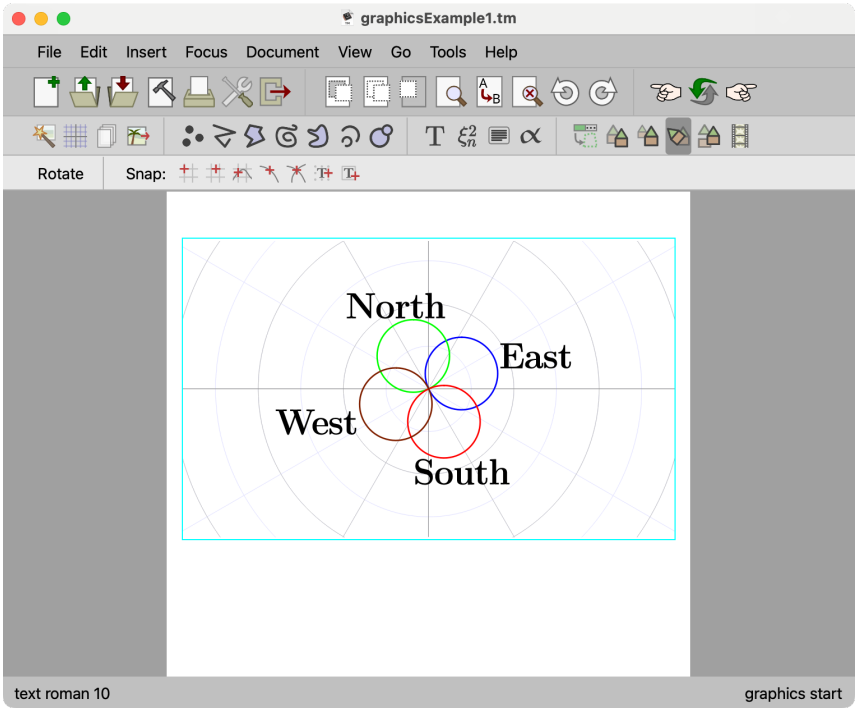
A left click on a group, in this mode, ungroups it again.

1.3.11 Resize and rotate the whole

Finally, to make the whole drawing smaller, select the Insert→Resize objects mode (icon), select the group with a right click if it is not selected yet, click with the left mouse button to start, move the mouse, and click again with the left mouse button when the size is right. We get for example:



To rotate the drawing, select the Insert→Rotate objects mode (icon) and proceed in the same way. The text boxes move with the rotation, but their text stays horizontal:



1.3.12 Change the geometry of the picture

You can change the size of the picture with Insert→Geometry→Size and Insert→Geometry→Resize, or with the keys ⌥←, ⌥→, ⌥↑ and ⌥↓; with Insert→Geometry→Crop→Crop, the picture is cropped automatically to the size of its contents (plus a padding). Insert→Geometry→Alignment sets the

vertical alignment of the picture with respect to the text. You can also zoom in and out of the whole picture with + and -, and shift the drawing with the arrow keys (when the picture is not cropped automatically).

2 Grids

The grid is set in the Insert→Grid menu (or with the icon ▦).

Action	*Behaviour*
Insert→Grid→Show grid	Show or hide the grid (also #)
Insert→Grid→Type	Set the type of grid: no grid, Cartesian, polar, logarithmic or notebook
Insert→Grid→Center	Set the center of the grid
Insert→Grid→Unit length	Set the unit length
Insert→Grid→Number of polar steps	Set the number of rays (polar grids only)
Insert→Grid→Logarithmic base	Set the base (logarithmic grids only)
Insert→Grid→Color of the axes	Set the color of the axes
Insert→Grid→Color of the units	Set the color of the unit lines
Insert→Grid→Show subunits	Show or hide the subunit lines
Insert→Grid→Color of the subunits	Set the color of the subunit lines
Insert→Grid→Number of subunit steps	Set the number of subunits per unit (also ^2, ^3...)
Focus→Snap	Choose what the points snap to: grid points, grid lines, curve points, intersections, text corners and borders; the snap distance

Table 1. The grid.

Older versions of TEXMACS had a separate, invisible *edit grid*; it is replaced by the snapping options of Focus→Snap.

3 Curves

3.1 Insertion modes

Action	*Behaviour*
Insert→Point	Insert points
Insert→Line	Insert broken lines
Insert→Polygon	Insert polygons
Insert→Curve	Insert splines, smooth curves or Bezier curves
Insert→Closed curve	Insert closed splines, smooth curves or Bezier curves
Insert→Arc	Insert arcs (through three points)
Insert→Circle	Insert circles (through three points)
Insert→Penscript, Insert→Calligraphy	Draw by hand

Table 2. The insertion modes for curves.

3.2 Operations

Action	*Behaviour*
Left click	Start a new curve, or add a point to the curve being drawn
Double left click	Add the last point and finish the curve
Right click	Remove the last point of the curve being drawn; otherwise, remove the control point under the mouse, like the middle button (with ⇧: the whole object)
Left drag and drop	Move the control point under the mouse, or insert a new point on the edge under the mouse
Middle click	Remove the control point under the mouse
⇧ and middle click	Remove the object under the mouse
⇥	Make the next control point or object under the mouse the current one
⌫, ⌦	Remove the control point under the mouse, like the middle button
^⇞, ^⇟	Move the current object closer to or farther from the eye
⌘Z	Undo (as everywhere in TEXMACS)

Table 3. Editing curves.

4 Text boxes

Action	*Behaviour*
Insert→Text	Insert text boxes
Insert→Mathematics	Insert mathematical formulas
Insert→Long text	Insert boxes with several paragraphs
Left click outside the boxes	Insert a new box and put the text cursor into it
Left click on a box	Put the text cursor into the box
Left drag and drop	Move the box under the mouse
Middle click, right click	Remove the box under the mouse

Table 4. Text boxes.

5 Properties of the objects

Action	*Behaviour*
Insert→Set properties	Enter the mode for changing the properties of existing objects; in this mode, the values chosen below apply at once to the selected objects
Focus→Color	Set the current color
Focus→Fill color	Set the current fill color
Focus→Opacity	Set the current opacity
Focus→Point style, Focus→Point size	Set the style and the size of points
Focus→Line width	Set the current line width
Focus→Line dashes	Set the current dash pattern
Focus→Line arrows	Set the current arrows
Focus→Horizontal alignment	Set the current horizontal alignment of text boxes
Focus→Vertical alignment	Set the current vertical alignment of text boxes
Left click (in Set properties mode)	Select the object under the mouse (a right click adds it to the selection)
↩	Apply the current properties to the object under the mouse
⇧↩	Make the properties of the object under the mouse the current ones

Table 5. Properties.

The menus show only the properties which make sense in the current mode: for instance, the alignments are in the Focus menu of the text modes.

6 Selecting and grouping the objects

Action	*Behaviour*
Insert→Group/ungroup	Enter the group/ungroup mode (the selection actions below are also available in the other group modes)
Right click	Add the object under the mouse to the selection, or remove it
Right drag and drop	Select all the objects inside a rectangle
, middle click	Forget the current selection
Left click	Group the selected objects, or ungroup the selected group
Edit→Copy, Edit→Cut	Copy or cut the selected objects
Edit→Paste	Paste the objects, at the place of the copied ones; they are selected
⇧ and middle click	Remove the selected objects (if nothing is selected, the object under the mouse)

Table 6. Selecting and grouping.

7 Transforming the objects

Action	*Behaviour*
Insert→Move objects	Enter the move mode
Insert→Resize objects	Enter the resize mode
Insert→Rotate objects	Enter the rotate mode
Left click	Start the operation on the selected objects, or on the object under the mouse if nothing is selected; a second left click finishes it
Left drag and drop	Perform the operation in one move

Table 7. Transforming objects.

8 Miscellaneous

The following keys act on the whole picture when the cursor is inside it.

Action	*Behaviour*
+	Zoom in
-	Zoom out
1, 2...	Zoom by a factor 1, 2...
←, →	Shift the drawing to the right, to the left
↓, ↑	Shift the drawing up, down
C	Put the origin at the center
⌥←, ⌥→	Decrease, increase the width of the picture
⌥↓, ⌥↑	Increase, decrease the height of the picture
#	Show or hide the grid

Table 8. Keys acting on the whole picture.

9 Since 2005

The plans of the first version of this tutorial were "other kinds of curves (i.e. conics)" and "zooming and rotating the text boxes". Meanwhile TEXMACS gained smooth and Bezier curves, the opacity and fill of the objects, hand-drawn curves, long text boxes, arrows with text, overlays and animations of pictures, and the snapping of the points to the grid and to the existing objects.

Long live Free Software!

Appendix A Overview of the graphical markup

A.1 Graphical objects

- point: ⟨point|x|y⟩;
- line: ⟨line|⟨point|x⟨rsub|1⟩|y⟨rsub|1⟩⟩|...|⟨point|x⟨rsub|n⟩|y⟨rsub|n⟩⟩⟩; cline (polygons), spline, cspline, smooth, csmooth, bezier, cbezier take their points in the same way;
- arc: ⟨arc|⟨point|x⟨rsub|1⟩|y⟨rsub|1⟩⟩|⟨point|x⟨rsub|2⟩|y⟨rsub|2⟩⟩|⟨point|x⟨rsub|3⟩|y⟨rsub|3⟩⟩⟩, and carc for circles;
- gr-group: ⟨gr-group|object⟨rsub|1⟩|...|object⟨rsub|n⟩⟩;
- text-at: ⟨text-at|markup|⟨point|x|y⟩⟩, and math-at and document-at for formulas and long texts.

A picture is a graphics tag containing these objects, usually inside a with which gives the frame (*gr-frame*), the size (*gr-geometry*), the grid (*gr-grid*) and the current properties of the editor (*gr-color*, *gr-mode*...).

A.2 Graphical properties

The properties are set with with around the objects:

- *color*, *fill-color*: red, blue, green, pastel red, #4040ff...;
- *opacity*: 0% to 100%;
- *point-style*: disk, round, square...; *point-size*: 2.5ln...;
- *line-width*: 1ln, 2ln, 0.01cm...;
- *dash-style*: a pattern of 0 and 1, like 11100; *dash-style-unit*: 5ln...;
- *arrow-begin*, *arrow-end*: <, >, |>, >>, o...;
- *text-at-halign*: left, center, right;
- *text-at-valign*: bottom, base, axis, center, top.

For instance, the blue circle of our example is

```
⟨with|color|blue|line-width|2ln|⟨carc|⟨point|0|0⟩|⟨point|0.5|-0.866⟩|⟨point|0.5|0.866⟩⟩⟩
```

Appendix B Graphical output using Scheme

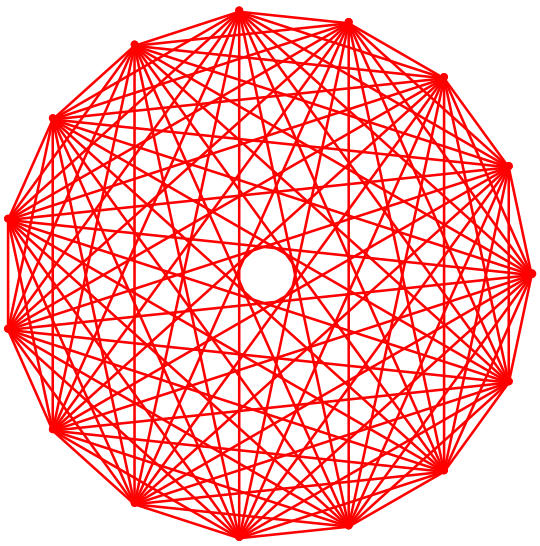
In this appendix, we show a Scheme function which draws the lines linking together a set of points on a circle, each of the points being linked with all the others. Its parameters are the radius of the circle, the number of points, the color and the line width of the lines. The function returns a picture, as $\TeX_{\text{MACS}}$ markup in Scheme form.

```
(tm-define (rose r n col lw)
  (:secure #t)
  (let* ((n (string->number (tree->string n)))
         (rad (/ (length-decode (tree->string r))
                 (length-decode "1cm")))
         (pi (acos -1))
         (pts (map (lambda (i)
                     (let ((t (/ (* 2 pi i) n)))
                       `(point ,(number->string (* rad (cos t)))
                               ,(number->string (* rad (sin t))))))
                   (iota n)))
         (lines (append-map (lambda (p)
                              (map (lambda (q) `(line ,p ,q))
                                   (cdr (member p pts))))
                            pts))
         (size (string-append (number->string (* 2 (+ rad 0.2))) "cm")))
    `(with "gr-geometry" (tuple "geometry" ,size ,size "center")
           "color" ,(tree->string col)
           "line-width" ,(tree->string lw)
           "fill-color" ,(tree->string col)
       (graphics "" ,@lines ,@pts))))
```

Put this code in your file `my-init-texmacs.scm` (in the folder `progs` of your TEXMACS home folder, `~/.TeXmacs` on LINUX and MACOS, `%APPDATA%\TeXmacs` on WINDOWS), or evaluate it in a SCHEME session (Insert→Session→Scheme). The extern primitive embeds the function in a macro:

```
⟨assign|rose|⟨macro|r|st|col|lw|⟨extern|rose|r|st|col|lw⟩⟩⟩
```

Using an instance of this macro, for example ⟨rose|3.5cm|15|red|2ln⟩, we get the following drawing:



That's all folks!